

Venkat Ramana, vice president, Hinditron Infosystems, a premier business and technical support partner of Boston Ltd in India, says, "With energy-efficient computing being the major focus area in the realm of high-performance computing (HPC), it is time to explore alternatives to the current industry-standard processors. In this context, presently, the ARM architecture seems to be the most ideal candidate. ARM processors are available from many vendors, and have the required economies of scale because of their sheer volume and wide deployment in the mobile market."

Calxeda's EnergyCore is one of the first ARM-powered SoCs that has been specifically designed to address the server market. Ganesan feels that Calxeda is uniquely placed because the well-funded start-up focuses solely on the server industry, while other ARM silicon partners that are entering or intend to enter the server market are engaged in other application areas to generate revenue. By far, EnergyCore is the most highly-integrated server processor on the market. In addition to including networking and storage interfaces on-chip, the chip uses a special fabric to remove the need for traditional top-of-rack switches and external baseband management controller devices, significantly reducing system power and cost. In terms of applications also, it is specifically tuned to the needs of data-centre applications, such as Web hosting, storage servers, data analytics and content streaming.

Boston's Viridis was one of the first ARM-powered server platforms to enter the market, and the first to be available in India. It was also the first to work so closely with Calxeda and make full use of the EnergyCore SoC.

Viridis—a quick overview

"Boston was the first company in the world to launch server-grade products based on ARM. We had a deep engagement with Calxeda in bringing our solutions to the market, and Viridis is the result. The Viridis system is designed from the ground up to deal with the requirements of scale-out data-centre workloads, with the internal fabric capable of scaling to over 4000 nodes today (without the need to go to a top-of-rack switch) and future releases that will scale to hundreds of thousands of nodes,"



Venkat Ramana, vice president,
Hinditron Infosystems



Manoj Nayee, Manoj Nayee,
managing director, Boston Ltd, UK

says David Power, technical head of the HPC group, Boston Ltd, UK.

At the heart of Viridis is SoC technology, which integrates the central processing unit (CPU), networking and input-output (I/O) onto a single server chip. Viridis uses Calxeda's ARM-based EnergyCore SoCs, which consume less than 5W of power. These are interconnected by a special highly scalable fabric that offers extremely low latency

high-bandwidth communication between all servers. This fabric is coupled with an intelligent management controller for power and fabric optimisation and management, for energy-proportional computing. The Viridis platform has five core chips, of which one is reserved for the management engine; only four cores are exposed to the operating system. Thus, the management and fabric do not interfere with the host performance. The disk I/O allows for up to 5 SATA devices to be populated per host, for users with data-intensive requirements.

Manoj Nayee, managing director, Boston Ltd, UK, says, "Using the very same CPU technology that powers many of today's mobile consumer devices, the Viridis offers around 90 per cent improvements in energy efficiency and density. These building blocks of high-end computing are set to radically change the economics of large-scale data centres, sparking innovation in emerging fields such as cloud computing, data modelling and analysis (often called 'Big Data'), scientific research and media streaming."

Low-power engineering

According to Power, the platform had to undergo a lot of engineering to get the product down to the power levels seen today. These include: silicon optimisations, EnergyCard optimisations, power-optimised fabric settings, optimised power supply, baffles to improve airflow, memory/RAM settings, and fan settings.

"ARM was always the technology we had planned on using for our Viridis product. It had to be ARM, as we wanted to bring a really disruptive green technology to the server market. Power consumption (and the cost!) was becoming a real headache for a lot of our customers, and ARM was one of the leading low-power chips in the market. We have a very close working relationship with ARM, and their roadmap is very exciting. We have also been engaged with Canonical (Ubuntu) and Red Hat/Fedora, who are heavily involved in porting their distributions to the ARM platform; I'm delighted at the progress that has been made in this space, particularly to provide users with a familiar working environment (both for development and administration). This combination made the choice of ARM a very compelling proposition. You also have communities and organisations such as Linaro (<http://www.linaro.org/>), which are investing a lot of



David Power, technical head,
HPC group, Boston Ltd, UK